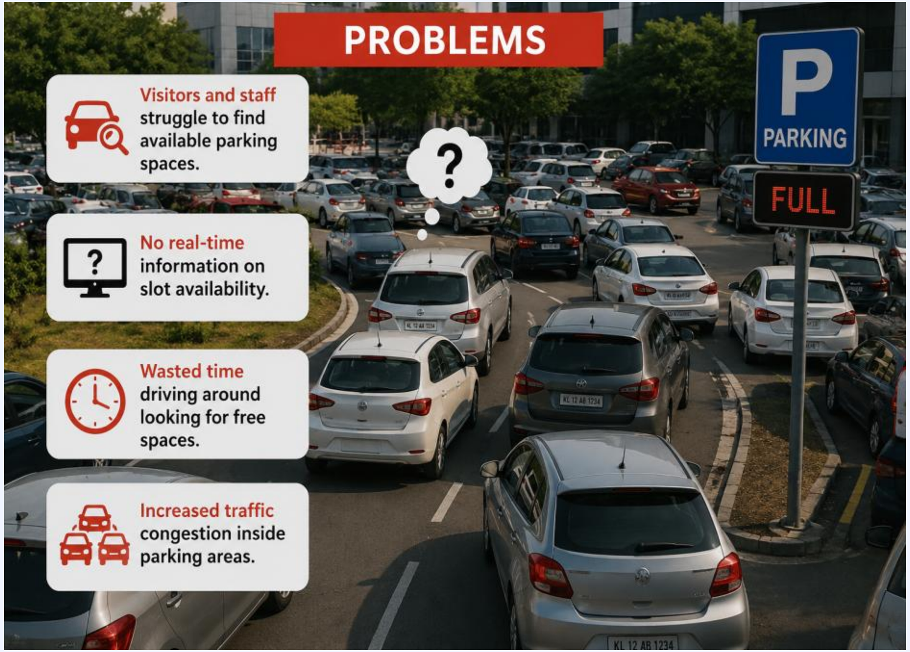
VIDYA NGO • TECHFEST 2026

SMART PARKING SYSTEM

Automated Slot Detection with IR Sensors & LCD Display

Trainer: Shruti Doke | Students: 4 | Components: Arduino UNO · 6× IR Sensors · Servo · LCD · Breadboard

PROBLEM STATEMENT



The Problem

- Visitors and staff struggle to find available parking spaces.
- No real-time information on slot availability.
- Wasted time driving around looking for free spaces.
- Increased traffic congestion inside parking areas.

This is especially critical for schools, hospitals, and malls with high footfall.

OUR SOLUTION

Slot Detection

6 IR sensors — one per slot — detect car presence in real time

LED Display

16×2 LCD shows available slots: 6→5→4→3→2→1 → FULL

Servo Barrier

Servo motor controls entry gate — opens when slots available

Together: a fully automated, low-cost smart parking system — no internet required!

HOW IT WORKS

1. Car enters parking area
2. IR sensor detects car
3. Arduino reads sensor data
4. Slot count updated
5. LCD shows available slots

LCD DISPLAY OUTPUT

Slots: 6	Slots: 5	Slots: 4	Slots: 3	Slots: 2	Slots: 1	FULL

← Available slots decrease as cars park. All 6 occupied = PARKING FULL

COMPONENTS USED

Arduino UNO

ATmega328P microcontroller — the brain of the project

IR Sensor × 6

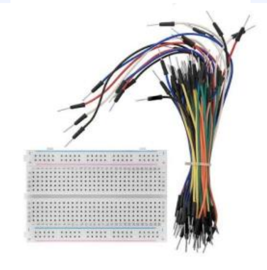
Detects car presence in each parking slot

Servo Motor × 1

Controls the entry/exit barrier gate

16×2 I²C LCD Display

Shows available slot count and PARKING FULL message



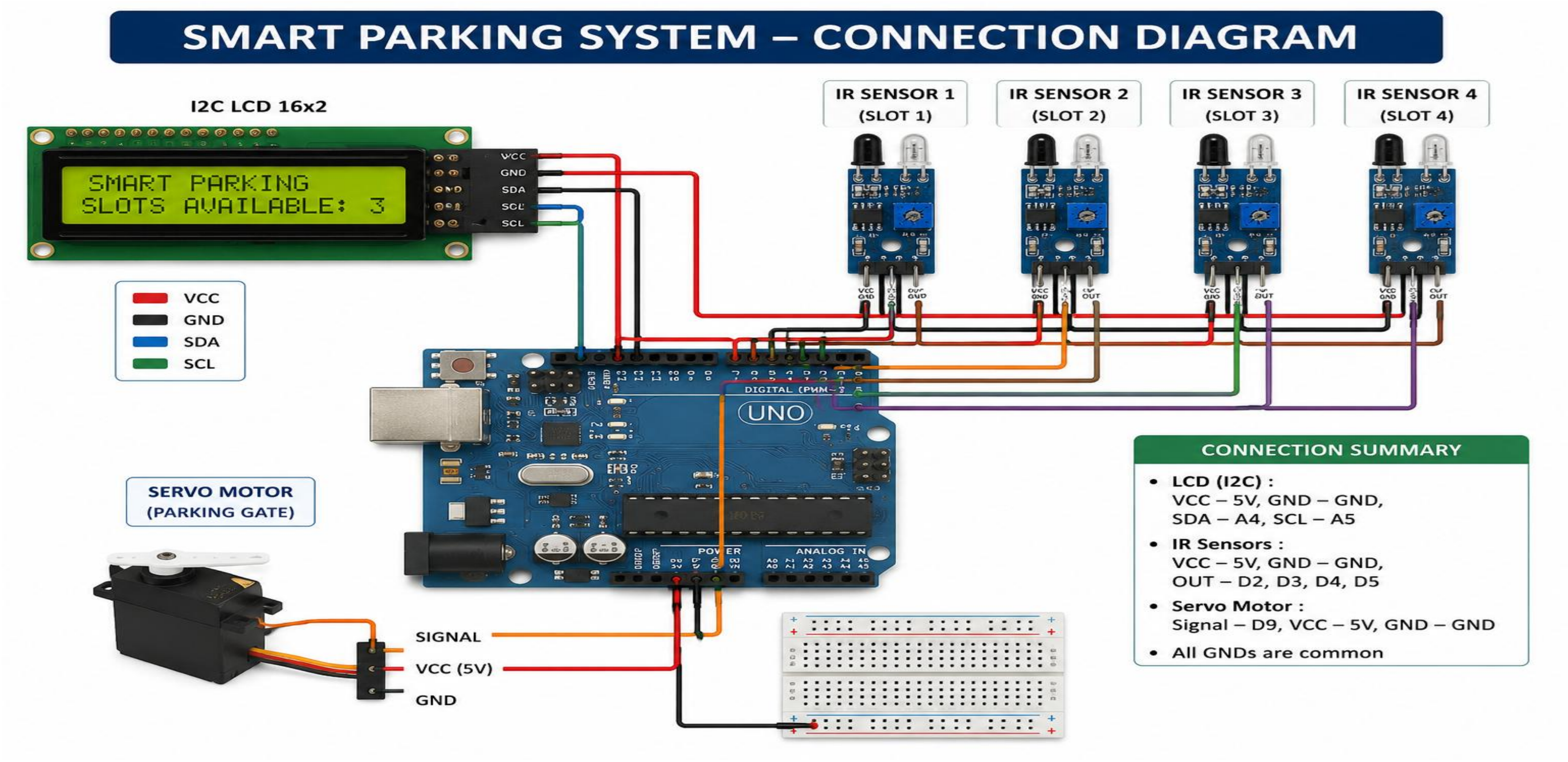
Breadboard + Wires

Connects all components without soldering

USB Cable

Powers the Arduino UNO from a laptop or USB wall adapter

CONNECTIONS



ADVANTAGES

Saves Time

Drivers know instantly which slot is free — no more driving around searching for a space.

Reduces Congestion

Less time spent circling means less traffic inside the parking area.

Organised Parking

Every slot is monitored. Parking becomes systematic and stress-free.

Low Cost & Simple

Built with Arduino + IR sensors — affordable and easy to replicate in any school lab.

No Internet Needed

Fully standalone system. Works offline with just USB power.

Easy to Maintain

Breadboard design makes it simple to replace or upgrade any component.

REAL LIFE APPLICATIONS

Shopping Malls

Thousands of visitors daily — real-time slot visibility prevents chaos at peak hours.

Hospitals

Patients and visitors need quick access. Smart parking reduces stress in emergencies.

Schools & Colleges

Staff and parent vehicles managed efficiently during drop-off and pick-up times.

Office Complexes

Employee parking managed automatically — no manual gate operator needed.

FUTURE SCOPE

Mobile App & Booking

Check available slots and reserve a spot in advance, right from your phone.

GPS Navigation

Guide drivers turn-by-turn directly to the nearest available parking slot.

Number Plate Recognition

Automatic vehicle entry and exit using camera-based plate detection (ANPR).

Cashless Payment

Pay parking fees seamlessly through UPI, QR codes, or cards.

EV Charging Stations

Dedicated charging slots to support the growing fleet of electric vehicles.

IoT Cloud & AI Monitoring

Monitor parking data remotely and predict peak-hour availability with AI.

THANK YOU!

Smart Parking System — Dr Babasaheb Ambedkar School

Trainer: Shruti Doke · 4 Students · Tinkering Project 2026